

Sector-Neutral Carry in a Synthetic Cross-Section

Disrupt Quant 2026 Challenge | Anya Mishra | Development 2021-01-01 to 2024-06-28 | Public validation 2024-07-01 to 2024-12-31

1. Hypothesis

Within a sector, assets with a higher **carry_score** earn a higher subsequent return than their sector peers. I read **carry_score** as a slowly varying holding yield: a characteristic that compensates an investor for bearing something unattractive about the asset, rather than a price-history effect. Two implications follow and both are testable. First, the premium should be a *relative-value* effect inside a peer group, since the compensation is only meaningful against comparable assets; the sector-average level of carry should carry no information. Second, because a yield accrues with time held, predictive power should *grow* with the forecast horizon rather than decay, which is the opposite of what a microstructure or liquidity-provision effect would do.

2. Evidence, including what failed

I scanned all 18 candidate features by daily cross-sectional Spearman rank IC against the return the engine actually pays a new position (open t+1 to close t+1), with the target cross-sectionally demeaned so that no signal is rewarded for market beta. One IC per session gives 910 quasi-independent draws.

Signal	mean IC	t-stat	Verdict
return_1d (reversal)	-0.0475	-6.54	Real, but untradeable - see Costs
carry_score	+0.0238	+3.42	Retained
return_5d (reversal)	-0.0177	-2.59	0.43 correlated with return_1d; redundant
return_20d (momentum)	+0.0129	+1.83	Rejected: quintile spread is 0.01 bps
14 others	< 0.01	< 1.4	No content (sentiment, flow, macro 1-3, ...)

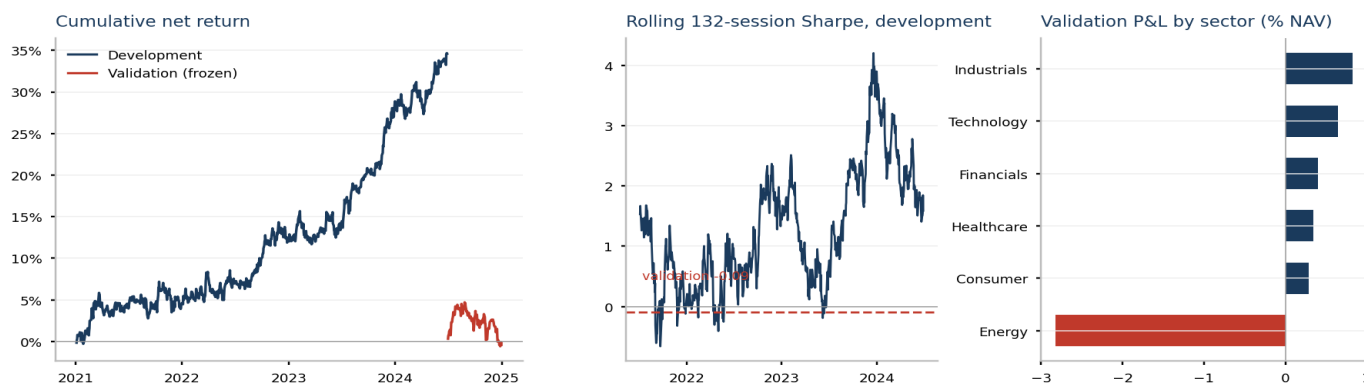
Both predictions of the hypothesis hold. Decomposing carry into a within-sector part and a sector-average part, **all** of the alpha sits in the within-sector component (IC +0.0270, t = +3.97) and none in the sector-average component (IC -0.0019, t = -0.24). And the IC rises monotonically with horizon: +0.018 at 1 day, +0.039 at 5, +0.063 at 20, +0.084 at 60. By contrast the reversal signal is entirely gone by t+2 (IC +0.009, t = 1.32), confirming the two effects are different in kind.

Conflicting evidence I am not discounting: the development IC strengthens materially through the sample (2021 t = 1.05, 2022 t = 1.84, 2023 t = 3.62, 2024 H1 t = 1.72). The full-sample t-statistic is therefore flattered by the middle of the period. I kept the signal because the sign is stable in every year and because the sector decomposition is a structural result, not a period-specific one, but the effect size is plainly uncertain.

3. From signal to positions

5-day mean of carry_score → **cross-sectional z-score** → **demean within sector** → **divide by realized_vol_20d** → **normalise to gross 1** → **scale to a 6% ex-ante annualised volatility target**. Ex-ante volatility is $w'\Sigma w$ from a 60-session sample covariance of asset returns, so leverage responds to the correlation structure rather than to individual volatilities alone. Leverage is clipped to [0.25, 2.0] and weights to $\pm 19\%$.

There are only three free parameters and none was optimised to a peak. The smoothing window was chosen at 5 sessions from a sensitivity sweep in which every setting from 1 to 90 sessions produced a development Sharpe between 1.30 and 1.57 - the choice is inside a flat region rather than on a spike. The covariance window (60) and the volatility target (6%) are risk-management settings, not return settings: across targets of 5%, 6% and 8% the Sharpe moves only between 1.59 and 1.61, because both alpha and cost scale close to linearly in position size. Selection experiments used an internal chronological split (fit 2021-2022, check 2023-2024 H1), never a shuffled split.



Left: cumulative net return, development and the once-only frozen validation run. Centre: the distribution of 132-session Sharpe ratios inside development, against the realised validation Sharpe. Right: validation attribution - five of six sectors positive.

4. Risk, sizing and diversification

The book is dollar-neutral and sector-neutral *by construction*, not by clipping: mean net exposure over development is 5×10^{-10} and the engine records **zero constraint-adjustment days** in both periods. This matters mechanically, because the engine responds to a breach by scaling the entire weight vector down rather than repairing the offending leg. Inverse-volatility weighting equalises risk rather than dollars across the 24 names; average gross exposure is 1.34 against a 2.0 limit, leaving headroom for the volatility scaler. Measured beta to the equal-weight universe is +0.0099 ($t = 0.70$), so essentially none of the return is directional. Maximum drawdown on development is -3.46% against 6.08% realised volatility.

Expected failure conditions: (i) carry stops being compensated and becomes a pure risk marker, in which case the premium disappears without warning; (ii) a shock that is common to the high-carry names across several sectors at once, which sector-neutrality does not hedge; (iii) a persistent widening of spreads, which would raise the cost floor under a signal whose gross edge is only ~2.3 bps/day. Financing and stock-borrow costs are excluded by the assessment convention. This flatters a long/short book more than a long-only one, and at an average gross of 1.34 a realistic borrow charge would reduce the reported return, though it would not change the sign of the result.

5. Costs and their effect on design

Costs are the reason this is a carry strategy and not a reversal strategy, and that was the single most consequential finding in the research. Reversal is by far the stronger statistical signal, but its cross-sectional score has day-over-day autocorrelation of **-0.04**: it reinvents itself every session, forcing ~146% of NAV of turnover per day. Gross alpha of 5.9 bps/day is then consumed by 7.1 bps/day of cost. Because spread dominates impact at this NAV, cost is close to linear in trade size, so scaling the book down cannot rescue it - both sides shrink together. I tested tail-only construction (top/bottom 2 to 8 names), spread-weighted cost-aware scoring, and no-trade bands of 2 to 6%; every variant remained negative net on both halves of the split. I report reversal as a rejected hypothesis rather than omitting it.

Carry's score, by contrast, has autocorrelation +0.994. It trades ~9.5% of NAV per day, costs ~0.5 bps/day against ~2.3 bps/day of gross alpha, and therefore keeps roughly 80% of its edge. The 5-day smoothing was adopted partly for this reason: it cut turnover by half while slightly *improving* Sharpe on both halves of the internal split.

Cost multiplier	1.0x	1.5x	2.0x	3.0x
Development Sharpe	1.38	1.29	1.19	1.00
Development ann. return	8.57%	7.94%	7.31%	6.06%
Validation Sharpe	-0.09	-0.20	-0.31	n/a

Sensitivity check: strategy decisions held fixed, NAV and trade sizes recomputed.

6. Development versus validation

	Sessions	Ann. return	Ann. vol	Sharpe	Max DD	Calmar	Turnover	Adj. days
Development	910	8.57%	6.08%	1.38	-3.46%	2.48	24.0x	0
Public validation	132	-0.78%	6.25%	-0.09	-5.06%	-0.15	27.6x	0

The strategy was frozen before validation was run, validation was run once, and nothing has been changed since. The result is flat to slightly negative and I am reporting it as it came out. Three diagnostics bear on whether it represents a broken signal or a small sample. First, the standard error of a Sharpe estimated over 132 sessions is $\sqrt{252/132} = 1.38$, so the validation figure sits roughly 1.1 standard errors below the development estimate; within development itself, 4.6% of overlapping 132-session windows produced a Sharpe at or below -0.09. Second, the signal went quiet rather than inverting: validation carry IC was -0.0135 with $t = -0.74$, statistically indistinguishable from zero. Third, attribution is highly concentrated - Energy contributed -2.82% while all five remaining sectors were positive, together contributing +2.51%. Energy was only 21% of average gross exposure, so this was signal failure in one sector rather than a sizing failure, which is precisely why I did not add a sector risk cap in response. Doing so would fit a control to a single observed loss and would convert validation into development.

7. Limitations and overfitting risk

The honest summary is that one signal out of eighteen survived, and the surviving signal is weak. Selecting the best of eighteen candidates inflates the apparent t-statistic even with an in-sample permutation test clearing at $p < 0.002$ (actual Sharpe +1.41 versus a null of -5.53 ± 0.57 from 500 within-day shuffles of carry), and a 20-day block bootstrap giving a 95% Sharpe interval of [+0.58, +2.26]. Those tests establish that the development result is not an artifact of the portfolio machinery; they cannot establish that the effect persists. The rising IC through development and the flat validation are jointly more consistent with a true Sharpe well below 1.38 - plausibly 0.3 to 0.7 - than with the development point estimate. Gross edge of ~2.3 bps/day is thin enough that a doubling of spreads or a halving of the premium would put the strategy at breakeven. Finally, carry_score has no economic definition I can verify, so I cannot rule out a relationship that is regime-switched by construction.

8. Single most useful next investigation

Determine whether the carry premium is *conditional* rather than constant. The specific test: estimate the within-sector carry IC in rolling 120-session windows across the full development sample and regress it on contemporaneous state variables - cross-sectional dispersion of carry, average realized volatility, and sector_dispersion. If the premium is systematically larger when carry dispersion is wide, that both explains the 2021-to-2023 strengthening and the 2024 H2 flatness, and yields an implementable rule: size the book in proportion to the dispersion of the signal and stand down when the cross-section is compressed. If no state variable predicts the IC, the conclusion is that the premium is constant but small and the development Sharpe was a favourable draw - which argues for running this at materially lower risk, or not at all.